

Trading Bot Types Reference

A comprehensive overview of common algorithmic trading bot architectures and strategies.

1. Trend Following Bots

Goal: Capture sustained directional market movements.

Examples

- Moving Average Crossover
- EMA Ribbon
- Donchian Channel Breakout
- Supertrend
- Turtle Trading
- ADX Trend
- MACD Trend
- Ichimoku Cloud
- Parabolic SAR
- ATR Trend Following

Best Market

- Strong bull or bear trends
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2. Mean Reversion Bots

Goal: Assume price returns to its historical average.

Examples

- RSI
- Bollinger Bands
- VWAP Reversion
- Z-Score
- Keltner Channels
- Williams %R
- Stochastic
- CCI

Best Market

- Sideways / ranging markets
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3. Grid Bots

Goal: Profit from market oscillations by placing multiple buy and sell orders.

Variants

- Spot Grid
- Futures Grid
- Long Grid
- Short Grid
- Neutral Grid
- Infinity Grid
- Dynamic Grid
- AI Grid
- Arithmetic Grid
- Geometric Grid

Best Market

- Range-bound markets
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4. DCA Bots

Goal: Average entry price over time.

Variants

- Fixed Interval DCA
 - Smart DCA
 - Indicator-triggered DCA
 - Martingale DCA
 - Anti-Martingale
 - Volatility DCA
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5. Arbitrage Bots

Goal: Exploit price differences across markets.

Types

- Exchange Arbitrage
- Triangular Arbitrage
- Spot-Futures Arbitrage
- Funding Arbitrage
- Cross-chain Arbitrage
- Cross-DEX Arbitrage

- Statistical Arbitrage
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6. Market Making Bots

Goal: Continuously provide liquidity and capture bid-ask spreads.

Variants

- Inventory-based
 - Avellaneda-Stoikov
 - Adaptive Spread
 - Hedged Market Maker
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7. Scalping Bots

Goal: Execute many small, quick trades.

Strategies

- Order Book Scalping
 - VWAP Scalping
 - DOM Scalping
 - Tick Scalping
 - Liquidity Scalping
 - Momentum Scalping
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8. Breakout Bots

Goal: Trade when price breaks out of consolidation.

Examples

- Opening Range Breakout
 - Donchian Breakout
 - ATR Breakout
 - Box Breakout
 - Volume Breakout
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9. Momentum Bots

Goal: Follow assets with increasing momentum.

Indicators

- ROC
 - Momentum Oscillator
 - RSI Momentum
 - MACD
 - ADX
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10. Swing Trading Bots

Goal: Hold positions for several days.

Signals

- EMA Pullbacks
 - Support & Resistance
 - Fibonacci Retracement
 - Trend Channels
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11. Position Trading Bots

Goal: Long-term investing based on macro trends.

Examples

- EMA 200
 - Economic Cycles
 - Bitcoin Halving
 - Factor Investing
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12. News Trading Bots

Goal: React automatically to news.

Sources

- Reuters
 - SEC Filings
 - Earnings Reports
 - Twitter/X
 - Crypto News
 - Federal Reserve announcements
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13. Sentiment Bots

Goal: Analyze crowd psychology.

Data Sources

- Twitter/X
 - Reddit
 - Telegram
 - Discord
 - Fear & Greed Index
 - Google Trends
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14. Order Flow Bots

Goal: Analyze actual buying and selling pressure.

Tools

- CVD
 - Footprint Charts
 - Delta
 - DOM
 - Volume Profile
 - Liquidity Heatmaps
 - Iceberg Detection
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15. Smart Money (ICT/SMC) Bots

Goal: Follow institutional-style price action concepts.

Concepts

- Order Blocks
 - Fair Value Gaps
 - Liquidity Sweeps
 - Break of Structure (BOS)
 - Change of Character (CHOCH)
 - Mitigation Blocks
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16. Statistical Trading Bots

Goal: Use mathematical models to identify trading opportunities.

Models

- Cointegration
 - Kalman Filter
 - PCA
 - ARIMA
 - GARCH
 - Bayesian Statistics
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17. Machine Learning Bots

Goal: Learn patterns from historical market data.

Algorithms

- Random Forest
 - XGBoost
 - LightGBM
 - CatBoost
 - Neural Networks
 - LSTM
 - Transformers
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18. Reinforcement Learning Bots

Goal: Learn optimal trading behavior through reward optimization.

Algorithms

- PPO
 - DQN
 - SAC
 - TD3
 - A3C
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19. High Frequency Trading (HFT)

Goal: Execute thousands of trades with extremely low latency.

Examples

- Latency Arbitrage
- Statistical HFT
- High-Speed Market Making
- Liquidity Sniping

20. Volatility Bots

Goal: Trade changes in market volatility.

Examples

- ATR Expansion
 - Bollinger Squeeze
 - Volatility Breakout
 - IV vs RV
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21. Options Bots

Strategies

- Covered Calls
 - Iron Condor
 - Straddle
 - Strangle
 - Butterfly
 - Calendar Spread
 - Gamma Scalping
 - Delta Hedging
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22. Funding Rate Bots

Goal: Exploit perpetual futures funding rates.

Examples

- Funding Capture
 - Funding Arbitrage
 - Premium Index Trading
 - Funding Prediction
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23. Liquidation Bots

Goal: Trade around forced liquidations.

Strategies

- Long Squeeze Detection

- Short Squeeze Detection
 - Liquidation Cascade Trading
 - Open Interest Analysis
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24. Copy Trading Bots

Variants

- Full Copy
 - Partial Copy
 - Risk-adjusted Copy
 - Portfolio Copy
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25. Portfolio Rebalancing Bots

Strategies

- Equal Weight
 - Risk Parity
 - Threshold Rebalancing
 - Volatility Targeting
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26. Hedging Bots

Types

- Delta Neutral
 - Spot/Futures Hedge
 - Beta Hedge
 - Cross Hedge
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27. Pairs Trading Bots

Examples

- BTC vs ETH
 - SOL vs ETH
 - Cointegrated Stocks
 - ETF Pairs
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28. Correlation Bots

Relationships

- BTC ↔ Nasdaq
 - Gold ↔ USD
 - Oil ↔ CAD
 - DXY ↔ Crypto
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29. Seasonal Bots

Examples

- Sell in May
 - Santa Rally
 - Weekend Effect
 - Month-End Effect
 - Bitcoin Halving Cycle
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30. Volume-Based Bots

Indicators

- OBV
 - MFI
 - CMF
 - VWAP
 - Volume Profile
 - Volume Spike Detection
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31. Pattern Recognition Bots

Chart Patterns

- Head & Shoulders
 - Double Top
 - Double Bottom
 - Triangles
 - Flags
 - Pennants
 - Wedges
 - Cup & Handle
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32. AI Hybrid Bots

Goal: Dynamically adapt strategies to market conditions.

Typical Architecture



Components

- Trend Detection
- Volatility Detection
- Strategy Selection
- Position Sizing
- Risk Management
- Trade Execution
- Performance Analytics

Summary Table

Bot Type	Best For
Trend Following	Strong trends
Mean Reversion	Sideways markets
Grid	Range-bound markets
DCA	Long-term accumulation
Arbitrage	Price inefficiencies
Market Making	Liquidity provision
Scalping	Short-term moves
Breakout	Volatility expansion
Momentum	Strong directional moves

Bot Type	Best For
Swing	Multi-day trades
Position	Long-term investing
News	Event-driven trading
Sentiment	Crowd psychology
Order Flow	Professional execution
Smart Money	Institutional concepts
Statistical	Quantitative models
Machine Learning	Pattern discovery
Reinforcement Learning	Adaptive learning
HFT	Ultra-low latency
Volatility	Volatility changes
Options	Derivatives strategies
Funding	Perpetual futures
Liquidation	Forced liquidations
Copy Trading	Following experts
Portfolio	Asset allocation
Hedging	Risk reduction
Pairs	Relative value
Correlation	Cross-market relationships
Seasonal	Calendar effects
Volume	Volume analysis
Pattern Recognition	Chart formations
AI Hybrid	Adaptive multi-strategy systems

Final Note

No single trading bot is universally profitable. The most resilient systems typically combine several of these approaches, using **market regime detection** to decide which strategy is most appropriate for current conditions. Combined with disciplined **risk management**, **position sizing**, and **execution**, this adaptive approach tends to be more robust than relying on any one strategy alone.